Global H5Nx sequences downloaded Dec. 10 2025

NCBI Virus (N_{sample}=19,449; N_{seq} =174,051) + **GISAID EpiFlu** (N_{sample}=42,415; N_{seq}=363,401)

Excluded passaged, vaccines, passaged, incomplete genome, and duplicated segments

Merged; Aligned; Gap-trimmed; Removed high N or low coverage; Removed identical genomes by host; Removed temporal outliers

Sample exclusion	NCBI	GISAID
Publication embargo (<3 months)	0	12
Vaccine	29	5
Passaged	4	62
Clones or lab derived	1173	167
Unknown host	135	7

Sequence exclusion	NCBI	GISAID
Replicate segments in sample	5651	23,985
<8 segments in sample	12,808	0

N_{sample} merged	N excl. N>10%, cover.<50%	N excl. identical genomes in host	N excl. temp outliers
61,491	59,889	41,427	41,366

Clean, de-duplicated H5Nx genomes
(N_{sample}=41,366, N_{seq}=330,928)

Segment	N > 10%	Cover. <50%	Temp. outliers: residuals	Temp. outliers: pendant edge
PB2	219	585	3	17
PB1	543	620	4	8
PA	147	571	4	10
HA	82	421	3	7
NP	24	412	4	3
NA	32	373	3	18
M	12	331	4	4
NS	35	345	4	4

Phylogenetic inference

Genotype and cluster assignment